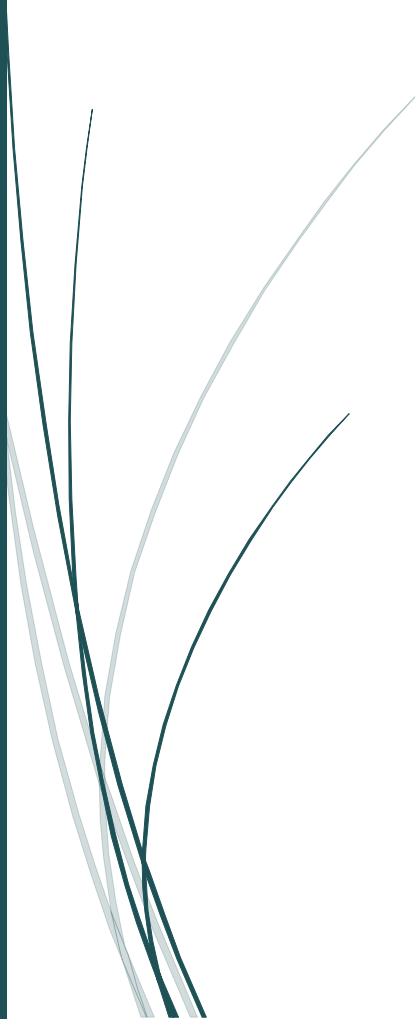
9/1/2026

N + - Network Fundamentals & Building Networks.

VLAN Concepts.



YAGNIK GHETIYA

TASK 1.

❖ On a Cisco Packet Tracer switch, create two VLANs named 'Engineering' (VLAN 10) and 'Marketing' (VLAN 20) using the CLI commands.

- **Creating VLANs on a Cisco Switch:**

  Commands used:

  Switch> enable
  Switch# configure terminal
  Switch(config)# vlan 10
  Switch(config-vlan)# name Engineering
  Switch(config-vlan)# exit
  Switch(config)# vlan 20
  Switch(config-vlan)# name Marketing
  Switch(config-vlan)# exit
  Switch(config)# end

  Result: Two VLANs were created successfully: VLAN 10 Engineering and VLAN 20 Marketing.

TASK 2.

❖ Assign ports FastEthernet0/1 and FastEthernet0/2 to VLAN 10, and ports FastEthernet0/3 and FastEthernet0/4 to VLAN 20 on your switch.

- **Assigning Switch Ports to VLANs:**

  Assign Ports to VLAN 10 (Engineering):

  Switch(config)# interface range fa0/1-2
  Switch(config-if-range)# switchport mode access
  Switch(config-if-range)# switchport access vlan 10

  Assign Ports to VLAN 20 (Marketing):

  Switch(config)# interface range fa0/3-4
  Switch(config-if-range)# switchport mode access
  Switch(config-if-range)# switchport access vlan 20

  Result: FastEthernet0/1 and FastEthernet0/2 are assigned to VLAN 10, while FastEthernet0/3 and FastEthernet0/4 are assigned to VLAN 20.

TASK 3.

❖ Write a short document (in a text file) listing the VLAN IDs, names, and the ports assigned to each VLAN from your configuration.

- **VLAN Configuration Details:**

  VLAN 10
  Name: Engineering
  Assigned Ports: FastEthernet0/1, FastEthernet0/2

  VLAN 20
  Name: Marketing
  Assigned Ports: FastEthernet0/3, FastEthernet0/4

TASK 4.

❖ Explain, in your own words, two benefits of using VLANs in a college campus network that has separate labs for coding and networking.

- **Two Benefits of VLANs in a College Campus Network:**

  - **Better Security:** VLANs separate coding lab computers from networking lab computers, helping restrict access to different sections of the network.

  - **Reduced Network Traffic:** Each VLAN has its own broadcast domain, reducing unnecessary traffic and helping devices communicate more efficiently.